SANGBAY
Gas Sensor

Combustible Gas Sensor

MJC4/2.8J

Catalytic Combustion Gas sensor

Small size | Low consumption | Long life | Low cost | High precision

Shanghai Sangbay Sensor Technology Co.,Ltd

1. Product Overview



MJC4/2.8J The catalytic combustion gas sensor is composed of a detection element and a compensation element, each of which is a platinum wire coil wrapped in an inert material. Gases capable of catalytic combustion will occur on the catalyst of the working element, causing a rise in the resistance of the detection element and a change in the output voltage of the bridge circuit. This change in value is linearly proportional to the concentration of flammable gases. Flammable gases will not undergo catalytic combustion reactions on the compensation element, therefore there will be no change in resistance. The compensation element mainly serves as a reference element and environmental temperature and humidity compensation function.

2. Application Field

1. Monitoring and alarming of various flammable gases such as methane andpropane in industrial production areas;

2.Gas monitoring and alarming for methane (primarily) in mining areas and underground corridors;

3.Fixed and portable inspection instruments for pipeline leakage monitoring.

3. Features

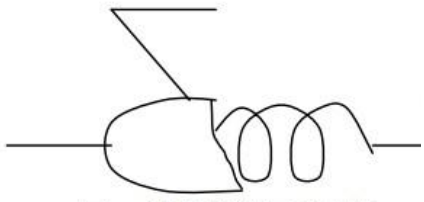
1. The output voltage of the bridge circuit is linear.
2. The response time is fast.
3. The sensor has good repeatability and selectivity.
4. The components work stably and reliability.
5. The sensor has excellent anti-H2S and anti-organic silicon

4. Technical Parameters

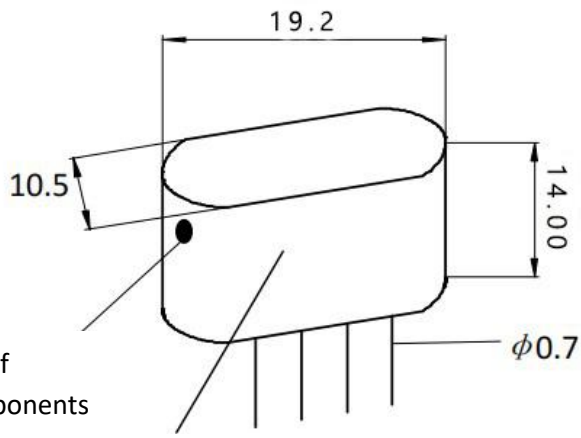
Item	Parameter
model	MJC4/2.8J
Detection principle	catalytic combustion
Detection gas	Combustible Gas
Working Voltage	2.8V（±0.1V）
Working Current	100mA（±10mA）
Detection range	0-100%LEL
Sensitivity	20-40mV / 1%CH_4
Zero-crossing voltage	±30mV
Resolution	1%LEL
Response Time	T_{90}≤30s
Recovery Time	T_{10}≤30s
Working Temp. Humidity	-20 ~ +60℃ ＜95%RH

Storage Temp. Humidity	-20 ~ +60℃ ＜95%RH
Zero Drift	≤±5% F.S /Month
Sensitivity Drift	≤±5% F.S /Month
Service Life	2 Years
ATEX	Ex da I Ma

Carriers and catalysts

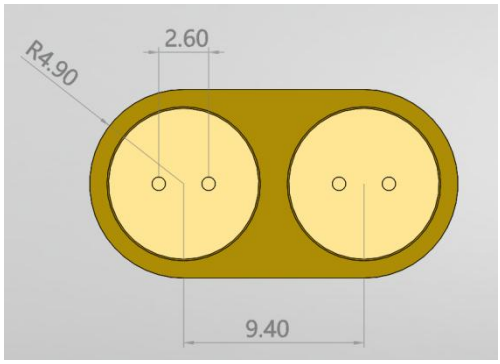


platinum wire



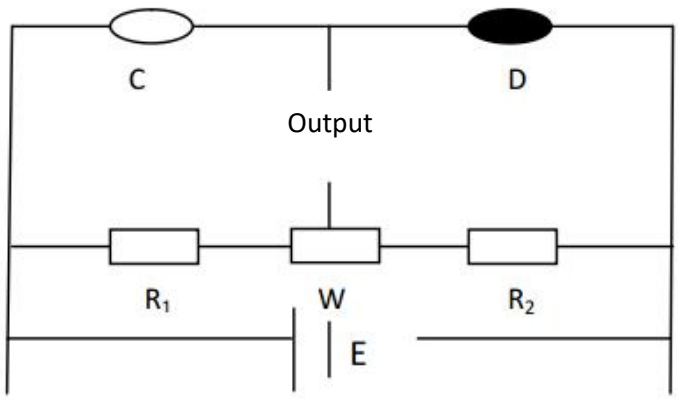
Identification of detection components

Powder metallurgy cover



(mm)

5. Basic Circuit



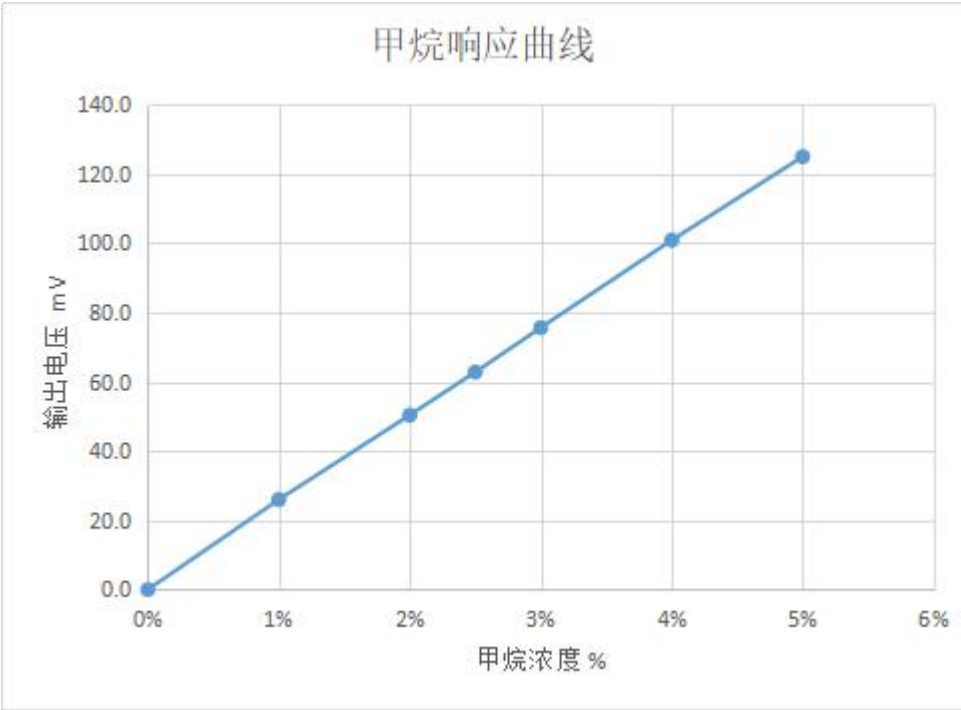
D detection element (Black)

C compensation element (white)

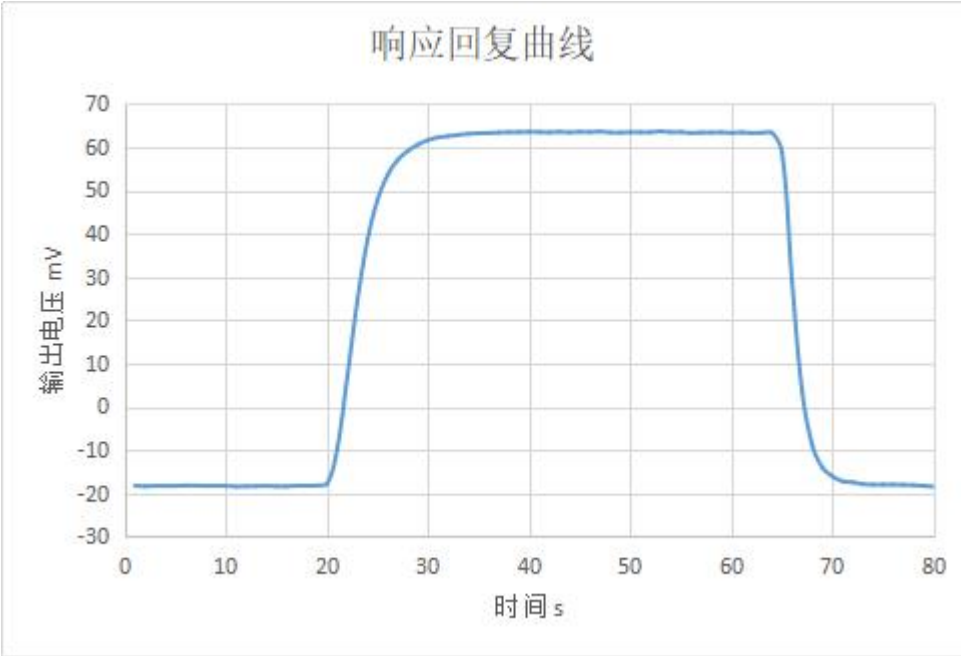
R1、R2=1K Bridge resistance

E Working power W Adjustable Potentiometer

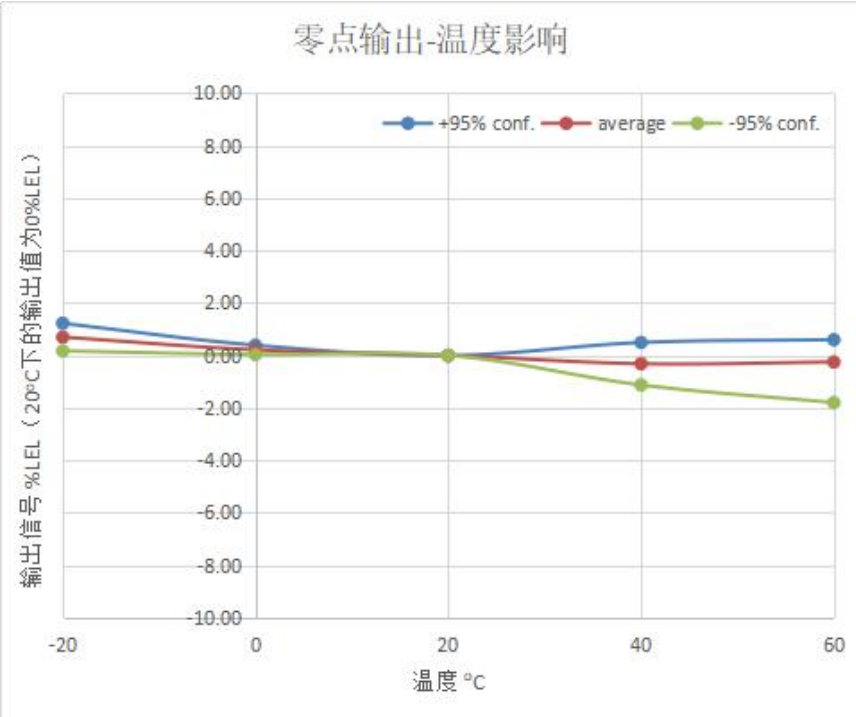
6. Characteristics

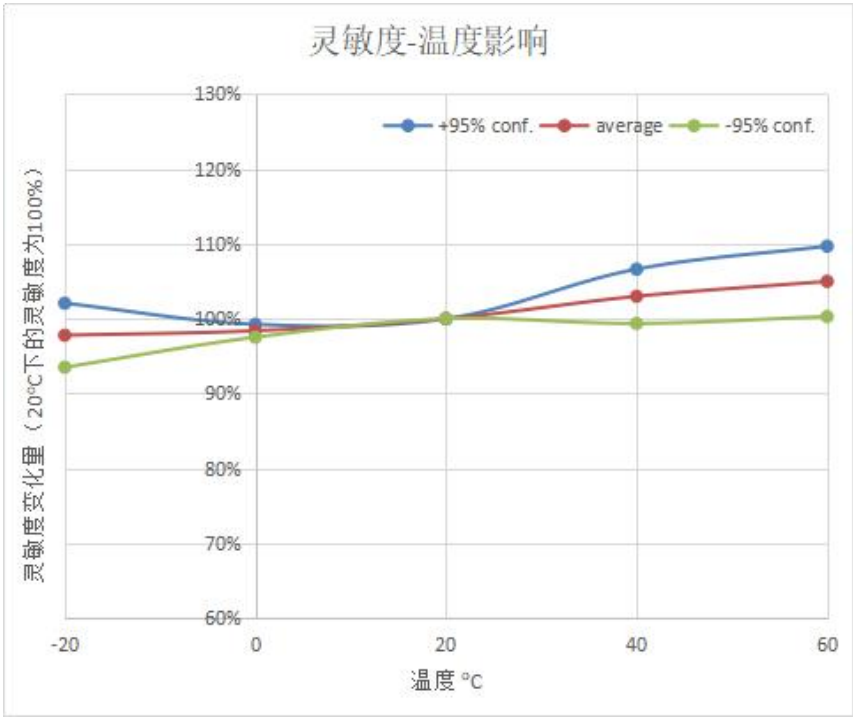


Typical sensitivity linear characteristic curve of sensors (CH4)



Sensor response recovery curve（2.5% CH4）





Sensor temperature characteristic curve

7. Cross Interference

MJC4/2.8JThe sensor responds to multiple types of flammable gases. The data in the table represents the relative sensitivity of some typical flammable gases at their corresponding concentrations of 100% LEL.

Molecular formula	100%LEL (vol%)	100%LEL Sensitivity
CH_4	5.0%	100
C_3H_8	2.1%	58
H_2	4.0%	45